

Changes in Colour Appearance of a Large Display in Various Surround Ambient Conditions

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Abstract: This article focuses on the change in colour appearance of a large display arising from various illumination conditions. Nine experimental phases were conducted according to different surround conditions including dark, indoor, and outdoor conditions. Each of the presented test colours was assessed by 10 observers using a magnitude estimation method. The surrounds used in all phases were divided into two groups: excluding and including veiling glare. Additionally, the effect due to different sizes of both stimuli and surround was investigated. Diverse visual effects were examined and reported. Finally, the visual colour appearance data were used to test the CIECAM02 colour appearance model. © 2010 Wiley Periodicals, Inc. *Col Res Appl*, 35, 200–212, 2010; Published online 14 January 2010 in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/col.20567

Key words: colour appearance; surround; lightness; colourfulness; hue; lightness contrast; CIECAM02

INTRODUCTION

The general trend in developing different types of display is to increase physical size together with a higher luminance and a larger colour gamut. These displays are seen under various viewing conditions such as inside a store for product promotion, in airports and train stations for the presentation of travel information, and increasingly for home theater. These viewing conditions typically include veiling glare¹ which is ambient light that is

reflected from the display. In such environments, consistent colour reproduction is important. There are many viewing parameters which affect the colour appearance of images on displays such as surround luminance level, display background and image size. The background is defined as the 10° field extending from the edge of a target stimulus in all or most directions and a surround is the area beyond this.² For self-luminous displays in real viewing environments, the surround is highly influential on appearance and it is for this reason that this parameter was studied here. When considering typical display viewing conditions, precise specification of the surround is often difficult because it includes a large range of luminances, veiling glare, inhomogeneous spatial configuration, different viewing geometries and viewing distances.³ Some colour-appearance experiments have been conducted using different types of display,^{4–6} although most were performed with a dark surround. Brainard *et al.*⁷ and Choh *et al.*⁸ investigated the effect of ambient illumination on the appearance of achromatic colours viewed on a CRT display. Their results showed an effect due to chromatic adaptation, but did not consider changes in the lightness and the colourfulness of the stimuli.

Various studies investigated the surround effect on perceived brightness and colourfulness. Stevens *et al.*⁹ found that perceived brightness increased as a power function of luminance with an exponent from 0.33 for a dark surround to a higher exponent for a bright surround. This is known as Stevens effect. Additionally, Bartleson and Breneman¹⁰ conducted a series of experiments to determine the relationship between perceived brightness and luminance by comparing five achromatic reflective photographic images with the corresponding projected transparencies in a dark surround. It was found that the shapes of the brightness function varied with not only luminance levels, but also surround conditions. Dark areas in an

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image appeared lighter under a dark surround than those under a bright surround. Further, a dark surround would reduce image contrast. This is known as Bartleson and Breneman effect. The aforementioned findings have been successfully applied in industries, i.e., a higher lightness contrast for a projected transparency film than that for a reflection print is required to produce an appearance match.¹¹

Hunt,¹² Pitt and Winter,¹³ and Breneman¹⁴ investigated whether surround influenced perceived colourfulness. They found that stimuli presented against a dark surround appeared less colourful than that against a bright surround. Hunt¹⁵ also found that background had an impact on colourfulness, but the extent of this effect depended on lightness; i.e., dark colours appeared more colourful against a darker background, whereas light colours appeared less colourful. Further investigation is needed to establish whether there is also a lightness dependency on colourfulness change due to the surround.

Recently, Liu and Fairchild¹⁶ investigated the change of perceived image contrast influenced by the change of surround luminance using colour images seen on a self-luminous display. In their experimental conditions, veiling glare was eliminated by placing the illuminating sources behind the display. Their results confirmed the Bartleson and Breneman effect. Only expert observers, however, tended to lower surround luminance when they perceived the image contrast to be too high, whereas nonexpert observers could not find any surround effect on image contrast. Their viewing conditions, however, may not be appropriate for real-world observation of images on displays where veiling glare is frequently caused by ambient light. Possible reasons for the discrepancy between expert and nonexpert observers include an insufficient range of stimuli in terms of contrast variation, attention being paid to different regions of interest in contrast assessment and differences in their adaptation.

This article investigates the change in colour appearance of a large display affected by surround conditions having a large range of luminances. The surround condi-



FIG. 1. An example for the surround excluding glare where observers cannot see the lighting sources directly because the lighting sources are covered by the diffusing cabinet and face the wall.



FIG. 2. An example for the surround including glare where observers can see the lighting sources affecting the displayed colours by adding glare.

tions were divided into two groups: excluding and including veiling glare from ambient lights. For the former, the illumination provided little veiling glare so the displayed colours were not changed but observers could perceive a surround of varying brightness. The two present experimental configurations used here are illustrated in Figs. 1 and 2. Figure 1 shows illumination located behind the display for the surround condition excluding glare. Figure 2 illustrates the surround including glare, a brighter surround introducing more veiling glare to the displayed colours. This simulates an exaggerated display viewing condition to illustrate the effect of real-world glare. Special attention was paid to reveal a true surround effect in the current experiment by introducing different levels of veiling glare by different luminance levels of the surround.

According to the International Commission on Illumination (CIE),¹⁷ surround can be defined by the surround ratio (S_R) which is calculated by dividing the luminance of a white in the surround area by the luminance of the reference white in the display area. Three surrounds were defined: average, dim and dark for $0.2 \leq S_R < 1$, $0 < S_R < 0.2$ and $S_R \approx 0$, respectively. All three surrounds were investigated in this study. In addition, a bright surround was added, which was defined as S_R being greater than 1.

The surround ratio (S_R) does not take into account the size of surround: the greater the viewing distance, the smaller the stimulus size (or the larger the size of the surround). In this study, the viewing distance was varied to investigate this effect. Finally, the performance of the CIECAM02 colour appearance model was also evaluated using the visual colour appearance data.

EXPERIMENTS

Psychophysical Experimental Setting

A 42-inch Plasma Display Panel (Samsung PPM42H3) was used in this study. Its pixel resolution was 1024×768 with a display area of $934 \times 532 \text{ mm}^2$. The long-term repeatability of this PDP was evaluated using 40 colours that were measured three times over a five-month period for all nine experimental phases, resulting in an

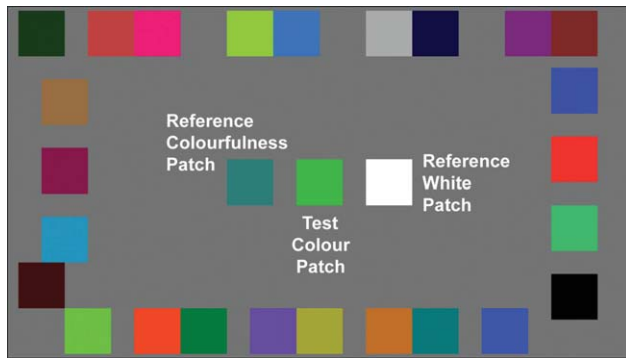


FIG. 3. Viewing pattern for colour appearance experiment.

average variation of $1.17 \Delta E_{ab}^*$ in the three measurements. Spatial uniformity was also estimated by measuring a displayed white colour at the center and also the surrounding eight positions. The average colour difference at eight locations compared to the center was $1.76 \Delta E_{ab}^*$. These results show that the spatial uniformity is slightly poorer than the long-term repeatability of this display. Three colour patches were needed for the colour appearance experiment: a reference colourfulness, a reference white and target samples. Therefore, four different configurations of the three colour patches were evaluated in order to find the best location, i.e., giving the minimum colour difference between any two of the three locations. It was found that the configuration in which the three colour patches were located adjacent horizontally at the display center was the best ($0.46 \Delta E_{ab}^*$).

Figure 3 shows the viewing pattern used in the experiment. All the colour patches had a size $70 \times 70 \text{ mm}^2$ subtending a visual field of 2° at a viewing distance of 2 m. The background in the display area had a viewing angle of $26.3^\circ \times 15.2^\circ$. There were 25 colours surrounding the edge of the pattern to form a complex viewing field. Forty test colours were selected to give a reasonable coverage of the colour gamut and lightness range. There were three colours in the middle of the display. The test colour was displayed in the center together with a reference white stimulus to the right and a reference colourfulness stimulus to the left. The spectral power distributions

of all the colours were measured using a Minolta CS1000 tele-spectroradiometer (TSR) and XYZ tristimulus values were calculated. The same viewing pattern as shown in Fig. 3 was utilized in the measurement of all the colours. Ten observers assessed each colour in terms of lightness, colourfulness and hue using the magnitude estimation technique⁴ in each of the nine experimental phases. In total, 10,800 estimations were made. For colourfulness scaling, a different reference colourfulness stimuli was employed in each phase, which was separately scaled in comparison with one physical sample viewed in D65 viewing booth. Hence, all colourfulness results assessed using different surrounds could be related each other. The anchor sample was a bluish red colour having an Natural Color System (NCS) specification of 3040-R20B.

Surround Conditions

Table I summarizes the viewing conditions of the nine experimental phases studied. The name of each phase was divided into four parts: surround (Drk, Dim, Avg or Brt), background (Gry or Blk), indoor or outdoor (In or Out) and stimulus sizes (2° or 0.6°). For example, DimGryIn0.6 means the experiment was conducted under a dim indoor surround condition with 0.6° stimuli assessed against a gray background.

Phases AvgGryOut2 and BrtGryOut2 were conducted externally to simulate the viewing conditions used for outdoor advertising. The shade behind a building was used to conduct the experiment to avoid direct sunshine and any nonuniform cloud shadow. Phases DimGryIn2 and DimGryIn0.6 were designed to simulate the viewing conditions found in an airport and a train station. DrkGryIn2 and DrkGryIn0.6 were designed to investigate the effect of stimuli sizes on perceived colour appearance.

As mentioned in the introduction, the surround ratio, S_R , was calculated from the ratio of the luminance of the white of the surround, to the luminance of the reference white of the display. The surround white was determined by measuring the luminance of a white ceramic tile that was illuminated by the light source used in each surround condition. For phases AvgGryOut2 and BrtGryOut2, a

TABLE I. Summary of the nine experimental phases conducted.

Phase	Surround	Surround white/Display white in cd m^{-2} (Surround ratio, S_R)	Y_b (Background)	Stimulus size ($^\circ$)
DrkGryIn2	Dark	0/253 (0)	19.3 (Gray)	2
BrtGryIn2	Bright	382/256 (1.48)	19.8 (Gray)	2
AvgGryIn2	Average	108/258 (0.42)	20.6 (Gray)	2
AvgGryOut2	Average	201/267 (0.78)	21.1 (Gray)	2
BrtGryOut2	Bright	282/274 (1.03)	24.4 (Gray)	2
DimGryIn2	Dim	48/256 (0.19)	20.6 (Gray)	2
DimGryIn0.6	Dim	48/257 (0.19)	20.6 (Gray)	0.6
DrkGryIn0.6	Dark	0/255 (0)	19.5 (Gray)	0.6
DrkBlkIn2	Dark	0/256 (0)	0.13 (Black)	2

Note that Y_b is background luminance factor that is the luminance ratio of background to the display white.

TABLE II. Classification of the experimental phases according to veiling glare.

Phase	Surround	Veiling glare (%)	Examination of surround effect dependent on existence of glare
DrkGryIn2	Dark	–	–
DimGryIn2	Dim	1.1	Excluding
BrtGryIn2	Bright	1.1	Excluding
AvgGryIn2	Average	2.0	Including
AvgGryOut2	Average	5.5	Including
BrtGryOut2	Bright	8.3	Including

white tile was placed in the center of the display, whereas in phases AvgGryIn2, BrtGryIn2, DimGryIn2, and DimGryIn0.6 each surround white was obtained by taking the average of the measurements of a piece of white paper which was placed at several locations to the right, left, above and below the display. The reason for measuring the surround white in the display center for the two outdoor phases was that their surrounds could be reasonably assumed to have uniform illumination due to the large distance between the light source (the sun) and the display. The luminance values for the reference white of the display were in the range 253–274 cd m⁻² dependent on the surrounds: their correlated colour temperatures (CCT) were all approximately 9,000 K. All test colour patches were measured using the TSR immediately after each observing session for Phases AvgGryOut2 and BrtGryOut2.

Table II classified the experimental phases used to examine colour appearance changes affected by the surrounds according to veiling glare. The veiling glare was estimated by measuring the luminance of the display's reference white with and without illuminated surrounds. To investigate the effect of surrounds excluding veiling glare, the visual data of DrkGryIn2 were compared with those of DimGryIn2 and BrtGryIn2. For Phase BrtGryIn2, an illuminator having 8 fluorescent lamps simulating D65 was located behind the display so that its light illuminated both the walls and the top of the display. Hence, it was thought that little flare was introduced onto the face of the display. For Phase DimGryIn2 (also for DimGryIn0.6), a lecture theater with evenly distributed fluorescent lamps in the ceiling was used. A uniform white screen and white walls were located behind the display and together formed the surround. The lamps were located far above the display, so the veiling glare was very small. However, this glare could not be removed completely for BrtGryIn2 and DimGryIn2, as seen in the second and third rows (1.1%) in Table II. This was because the light from the fluorescent lamps was reflected from the surrounding walls and thus was added to the displayed colours. Although the glare probably affected only the dark colours for BrtGryIn2 and DimGryIn2, the amount of glare was smaller compared with that for AvgGryIn2, AvgGryOut2, and BrtGryOut2. Thus, the visual data accumulated in BrtGryIn2 and DimGryIn2 were used to understand the effects of the surround excluding glare by comparing with the visual data in DrkGryIn2. How-

ever, a special attention was paid to separate the effects due to different luminance surrounds from those caused by the veiling glare added to dark colours. This was achieved by comparing the visual data and measured data simultaneously between DrkGryIn2 and each of BrtGryIn2 and DimGryIn2.

In the analysis of the effects of surround including/excluding glare, the visual results between the two phases being compared are plotted to reveal their difference. Where calculated using the measurement data of the displayed colours, the CIELAB L^* or C^* values of each pair of stimuli are also be plotted to indicate the amount of veiling glare. (In this case, the displayed colours were the same as those viewed by observers in each phase of the experiment.) CIELAB was used for simplicity and to reveal basic trends. To check the performance of the current recommended colour appearance model CIECAM02, the visual lightness, colourfulness and hue results are compared with the corresponding predictions made using the CIECAM02 in the last result section.

Observer Variation

The coefficient of variation, CV , defined in Equation (1), is a statistical measure to represent the agreement between two sets of data. When two appropriate sets of data are plotted, all the data points should lie on the 45° line and the value of CV should be zero for perfect agreement.

$$CV = \frac{100}{\bar{y}} \left[\sum (x_i - y_i)^2 / n \right]^{1/2} \quad (1)$$

where n is the number of samples and $\bar{y}$ is the mean value of the y data set.

In each phase, observers repeated their assessments for 10 of the 40 colours. The CV values were computed between the two repeated judgments to represent observer repeatability performance. In addition, the CV values between the individual and mean visual result were also calculated, which is an indicator of observer accuracy. The overall mean CV values for observer repeatability were 13, 21, and 6 for lightness, colourfulness, and hue, respectively. Those for observer accuracy were 15, 19, and 7, whereas for the LUTCHI data the values quoted are 13, 18, and 9, respectively.⁴

These results showed:

- that observers performed similarly in terms of accuracy and repeatability;
- that the performance of the present study was similar to that of the LUTCHI data;
- that hue results are most accurate among the three attributes and colourfulness is the most difficult to scale; and that
- the results also showed similar observer accuracy between the results including and excluding veiling glare, and between the indoor and outdoor conditions.

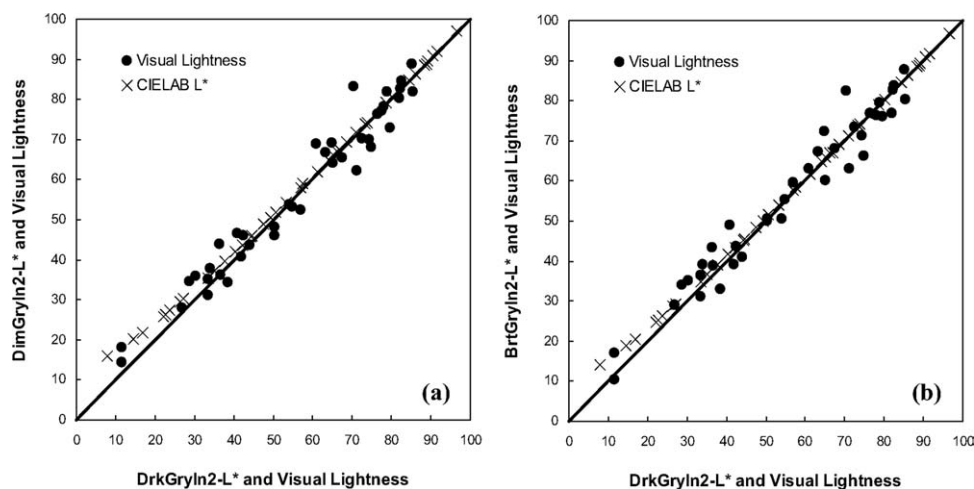


FIG. 4. Comparisons of the visual lightness (●) and the corresponding measured L^* data (x) of a series of test colours seen with a dark surround (DrkGryIn2), and the equivalent visual lightness and measured data (L^*) seen (a) with a dim surround (DimGryIn2) and (b) with a bright surround (BrtGryIn2). Also shown is a line at 45 degrees.

In the visual hue comparisons between Phase DrkGryIn2 (dark surround) and all the other phases, there was little disagreement, therefore only visual lightness and colourfulness results will be discussed.

COMPARISON OF VISUAL RESULTS

A comparison of visual results between any two different phases was made using the average of all observers' visual data.

Effect of the Surround Excluding Glare

This section explains the effects of the surround where veiling glare had little influence on the displayed colours. Observers made their visual assessments under varying-brightness surrounds. The visual results of BrtGryIn2 for a bright surround and DimGryIn2 for a dim surround were compared with those of DrkGryIn2 for a dark surround. Figures 4(a) and 4(b) show not only the visual lightness but also the CIELAB L^* values of the 40 colours assessed using the dim and bright surrounds plotted against the corresponding values in the dark surround. (Note that the L^* values were measured under the viewing condition of each particular phase. If the data points are located on the 45° line, this implies that there is no glare effect between the two phases studied.) The L^* values for very dark colours ($L^* < 30$) lie above the 45° lines, while the others are on the 45° lines in both figures. The higher L^* values for the very dark colours viewed in the bright (BrtGryIn2) and dim (DimGryIn2) surrounds were caused by reflections from surrounding light gray walls. There was not, therefore, complete elimination of ambient lights. Both figures reveal excellent agreement of the perceived lightness results between the dim and dark surrounds, and between the bright and dark surrounds, respectively.

The present result disagreed with the earlier finding of Bartleson and Breneman¹⁰ which states that dim and dark surrounds can make the whole picture look lighter, but this effect is greater for dark regions of the picture than for light regions. The likely reason for this discrepancy is the different experimental conditions between the present work and that of Bartleson and Breneman:

1. differences in the stimuli used in each of the two experiments—achromatic images (reflection prints viewed with illuminated surrounds and projected transparencies viewed with dark surrounds) for Bartleson and Breneman, while colour patches were used in the present work, and
2. differences in the size of stimuli used in each of the two experiments—the particular areas (which were not specified) in the printed images subtended 29° (H) × 29° (V) visual angle and in the transparency images subtended 15° – 20° (H) × 15° – 20° (V) for Bartleson and Breneman, whereas the colour patches (2° × 2°) against the gray background subtended 26° (H) × 15° (V) in the present work.

Considering the earlier, it is thought that the most experimental difference is in stimuli. Those used in Bartleson and Breneman's study were achromatic images and so lightness changes in colour stimuli arising from different surround conditions can be presumed to have the same tendency. As described in the introduction, it was found that bright surrounds could induce a colourfulness increase, whereas dark surrounds produced the contrary effect. Hence, the lightness contrast increase could be canceled by the colourfulness decrease in dark surrounds. This possible cancelation, previously pointed out by Hunt,¹⁸ led Fairchild¹⁹ to address the necessity of experiments verifying Bartleson and Breneman effect for colour image stimuli. To establish more accurate reasons for the

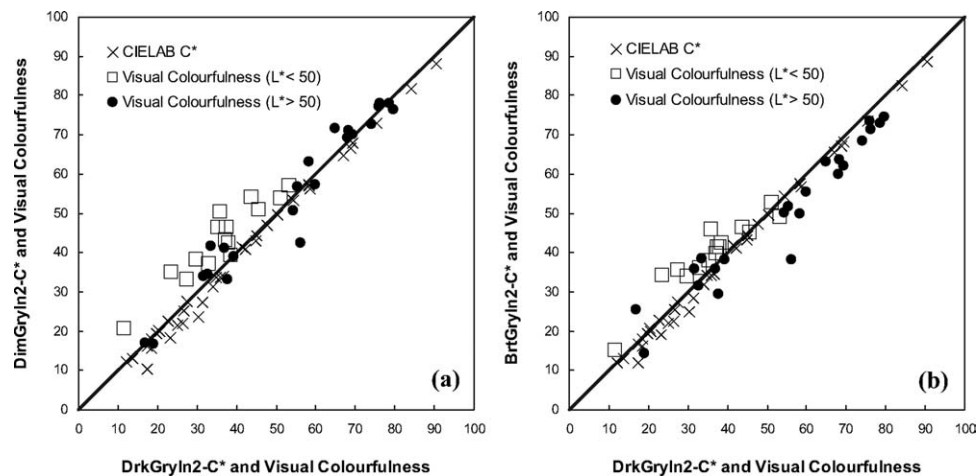


FIG. 5. Comparison of the visual colourfulness ($\square$ and $\bullet$) and the corresponding measured C^* data (x) of a series of test colours seen with a dark surround (DrkGryIn2) and the equivalent visual colourfulness and measured data (C^*) seen (a) with a dim surround (DimGryIn2) and (b) with a bright surround (BrGryIn2). The visual data symbolized by squares are dark colours having less than 50 of L^* . Also shown is a line at 45 degrees.

deviation between results from the current work and those from Bartleson and Breneman's study, further psychophysical experiments have been carried out using complex colour images. The results will be reported subsequently.²⁰

Figure 5 shows the comparisons of visual colourfulness and CIE LAB C^* values of the colours assessed in the same pairs of phases shown in Fig. 4. In each figure, the visual data points of dark colours (L^* less than 50) are drawn using square symbols in order to show the lightness dependency of colourfulness change. The C^* values for the very dark colours are located below the 45° lines in both figures [as the L^* values for those were located above the 45° lines in Figs. 4(a) and 4(b)]; this is due to the addition of light reflected from the surrounding walls for the dim (DimGryIn2) and bright (BrGryIn2) surrounds. This may be seen by comparing Fig. 5 with Figs. 8 and 11. In Fig. 8, the test colours were measured at 2 m and 7 m viewing distances under the same dim surround, and in Fig. 11 the test colours were measured against gray and black backgrounds under the same dark surround. All measured L^* and C^* data points in Figs. 8 and 11 are located on the 45° lines, indicating that the measured test colours were identical under the same surround and the display used provided consistent test stimuli. Therefore, it can be concluded that the ambient light reflected from the surrounding walls in DimGryIn2 and BrGryIn2 was added to the very dark colours, resulting in the C^* values positioned below the 45° lines in Figs. 5(a) and 5(b).

Despite the very dark colours having decreased values of C^* in the dim and bright surrounds, Figs. 5(a) and 5(b) show that the dark colours ($\square$) appear to be more colourful for the dim and bright surrounds than for the dark surround. Lighter colours ($\bullet$) appear less colourful under the bright surround than those under the dark surround, whereas the dim surround does not show this trend. This lightness-dependent colourfulness change arising from the

luminance change in surrounds excluding glare is a unique finding not reported elsewhere. To prove this phenomenon, more psychophysical experiments are required under environments free of veiling-glare and having a large range of luminance. Hunt,¹² Pitt and Winter,¹³ and Breneman¹⁴ found that colours look more colourful for bright surrounds than for dark surrounds, but they did not show different surround effects on colourfulness with respect to dark or light colours. It is thought that in their studies, attention was not paid to reveal lightness-dependent colourfulness changes. Hunt¹² attempted to explain the mechanisms for colourfulness decrease in dark surrounds: the responses of one colour (e.g., red) spill over into the nerve fibers connected predominantly to other colours (e.g., green and blue) for dark-adapted state (dark surrounds), leading to a loss of red colourfulness. However, there is currently no physiological evidence to support this suggested reason.²¹ Therefore, more psychophysical experiments are required to establish basic mechanisms in the human visual system to explain the surround effect on colour appearance.

Effect of the Surround Including Glare

This section describes effects of the surround, where veiling glare was included as illustrated in Fig. 2. The visual lightness and colourfulness results for phases including glare [AvgGryIn2 (average surround/indoor), AvgGryOut2 (average surround/outdoor) and BrGryOut2 (bright surround/outdoor)] are plotted against those of DrkGryIn2 (dark surround/indoor, without glare) in Figs. 6 and 7, respectively. The L^* and C^* values are also plotted to illustrate the degree of glare. An obvious glare effect can be found from the colour measurement result where, for a brighter surround, the L^* values are gradually increased towards darker colours (see Fig. 6) and there is a consistent decrease of C^* values (see Fig. 7).

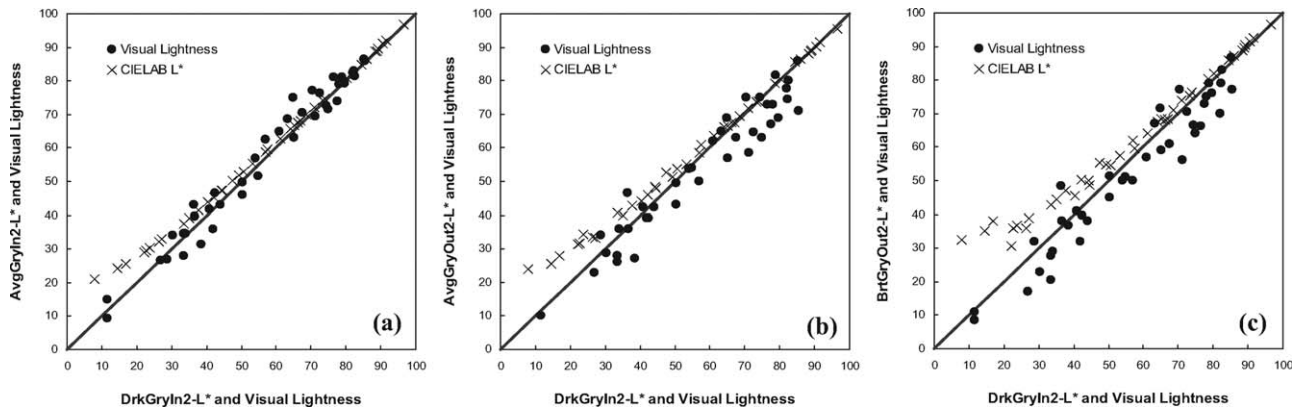


FIG. 6. Comparisons of the visual lightness (●) and the corresponding measured L^* data (x) of a series of test colours seen with a dark surround (DrkGryIn2), and the equivalent visual lightness and measured data (L^*) seen (a) with an average indoor surround (AvgGryIn2), (b) with an average outdoor surround (AvgGryOut2), and (c) with a bright outdoor surround (BrtGryOut2). Also shown is a line at 45 degrees.

For light colours (measured L^* greater than 70), there is little L^* difference between colours seen against the dark surround and each of the three illuminated surrounds. For the two outdoor surrounds, however, they appear consistently darker than for the dark surround [Figs. 6(b) and 6(c)], which results in a reduction in the perceived dynamic range of lightness as the surround brightness (including glare) increases. In general, it is held that the lightness dynamic range decreases in bright outdoor conditions, because dark colours are perceived to be lighter in bright outdoor conditions than with dark surrounds. However, the finding in Figs. 6(b) and 6(c) suggests that the dynamic range can also be reduced due to light colours looking darker.

For dark colours (measured L^* less than 50) in Fig. 6, a clear glare effect can be found: L^* values are higher against the three illuminated surrounds than against the dark surround. There is, however, little visual lightness difference between the average indoor and dark surrounds [Fig. 6(a)]. Also, there is a small decrease in visual lightness for the two outdoor surrounds compared with the

dark surround [Figs. 6(b) and 6(c)]. Thus, although the dark colours were measured as being lighter in the three illuminated surrounds than in the dark surround, observers still perceived these to be at least as dark as those in the dark surround. One cause of this could be that observers unconsciously discounted veiling glare in their judgment of lightness. It is also possible that their perception of lightness was influenced by their memory of other test colours presented during the same experimental session. In the previous paragraph, it was mentioned that the light colours appeared slightly darker in the two outdoor surrounds [Figs. 6(b) and 6(c)] than in the dark surround. This could influence the lightness assessment of the dark colours as being darker, despite an increase in the measured L^* values. For this reason, and also because brightness is an absolute attribute, it needs to be assessed concurrently with lightness under a wide range of ambient lighting levels.

Figure 7 represents the visual colourfulness data of dark colours (L^* less than 50) using squares. There is no lightness-dependent colourfulness change, unlike the

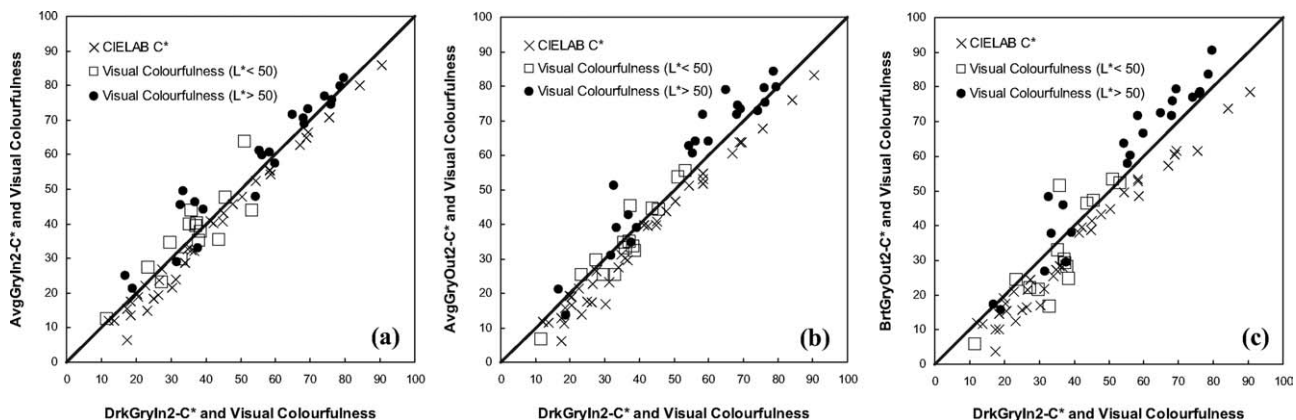


FIG. 7. Comparisons of the visual colourfulness (□ and ●) and the corresponding measured C^* data (x) of a series of test colours seen with a dark surround (DrkGryIn2), and the equivalent visual colourfulness and measured data (C^*) seen (a) with an average indoor surround (AvgGryIn2), (b) with an average outdoor surround (AvgGryOut2), and (c) with a bright outdoor surround (BrtGryOut2). Also shown is a line at 45 degrees.

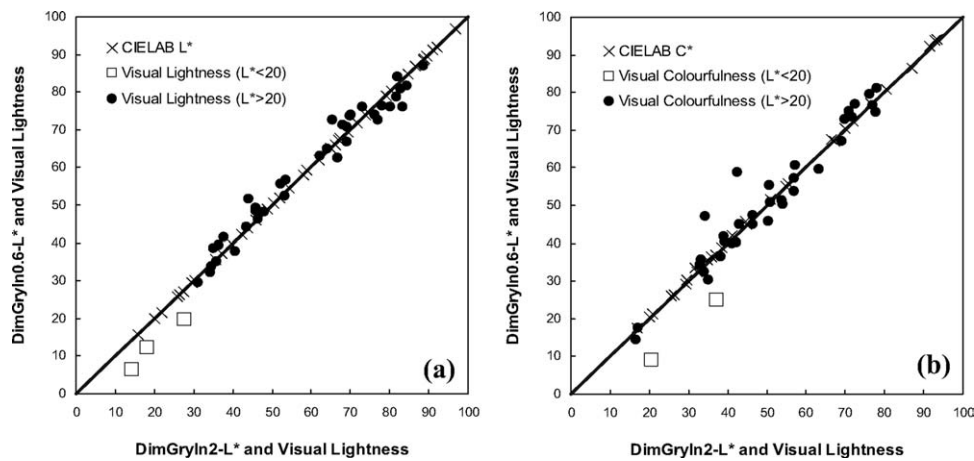


FIG. 8. Comparison of (a) the visual lightness ($\square$ and $\bullet$) and the corresponding measured L^* data (x), and (b) the visual colourfulness ($\square$ and $\bullet$) and the corresponding measured C^* data (x) of a series of test colours seen for DimGryIn2, and the equivalent data seen for DimGryIn 0.6, respectively. The visual data symbolized by squares are very dark colours having less than 20 of L^* . Also shown is a line at 45 degrees.

result for the surround excluding glare in the previous section. All C^* data points are below the 45° lines in Figs. 7(a)–7(c), indicating higher C^* values in the dark surround than in the three illuminated surrounds due to glare. Figure 7(a) shows, however, that colours appear slightly more colourful under the average indoor surround than under the dark surround. In Fig. 7(c), it can be seen that dark colours close to neutral appear less colourful; but medium and high chromatic colours appear more colourful under the bright outdoor surround compared to those under the dark surround. The cause of this result is that when observers estimated colourfulness for each test sample, they considered both the colourfulness of the reference colourfulness sample and that of the other colours previously presented. As expected, dark near-neutral colours tended to appear less colourful as brightness and glare increased from average indoor [Fig. 7(a)] to the bright outdoor surrounds [Fig. 7(c)]. Conversely, medium and high chromatic colours tended to appear more colourful.

The following visual phenomena can be established from the aforementioned results. Bright surrounds which include glare can have a large impact on perceived appearance. When an image is seen under bright outdoor conditions, highlight areas appear darker due to a visual lightness-reduction for light colours. Shadow-detail will also be lost due to a reduction in colourfulness perception for near-neutral colours. This leads to a poor discrimination of shadow-detail and darker appearance for highlight regions.

The Effect of Changes in Both Stimulus and Surround Size

To investigate the effect of stimulus size on visual colour appearance, the visual results for stimuli at two different angular sizes (2° and 0.6°), corresponding to viewing distances of 2 m and 7 m, respectively, were compared.

No obvious difference was found under the dark surround condition; however, a very small difference was seen under the dim surround. Figures 8(a) and 8(b) plot the visual lightness and colourfulness results, respectively, of the dim surround at a 7 m viewing distance (Phase DimGryIn0.6) against those of the dim surround at a 2 m viewing distance (Phase DimGryIn2). The very dark colours ($\square$) appear darker and less colourful for the 0.6° stimuli than for the 2° stimuli. This small effect could be caused by the surround size difference, as illustrated by Fig. 9. When observers experienced a larger and brighter surround (or smaller colour stimuli viewed at a longer distance), dark colours appeared darker and less colourful. This is thought to be one of the reasons why colour appearance is affected by changes in stimulus and surround sizes. To establish and quantify this effect, more psychophysical experiments are required.

Comparison of the Bright Surrounds with and without Glare

This section investigates the glare effect on colour appearance. Figures 10(a) and 10(b) plot the visual lightness and colourfulness results, respectively, of the bright surround including glare (Phase BrtGryOut2) against those of the bright surround excluding glare (Phase BrtGryIn2). The effect is very clear that glare makes all colours appear slightly darker. The reason for this phenomenon was given in “Effect of the Surround Excluding Glare” and “Effect of the Surround Including Glare” sections: compared to the visual lightness of test colours in the dark surround, little difference was found against the bright surround excluding glare, whereas a darker appearance was evident for light colours against the bright surround including glare. Therefore, it is proposed that the reduced visual lightness for light colours has arisen from the existence of glare in the bright surround. For the effect on visual colourfulness, glare causes low chromatic

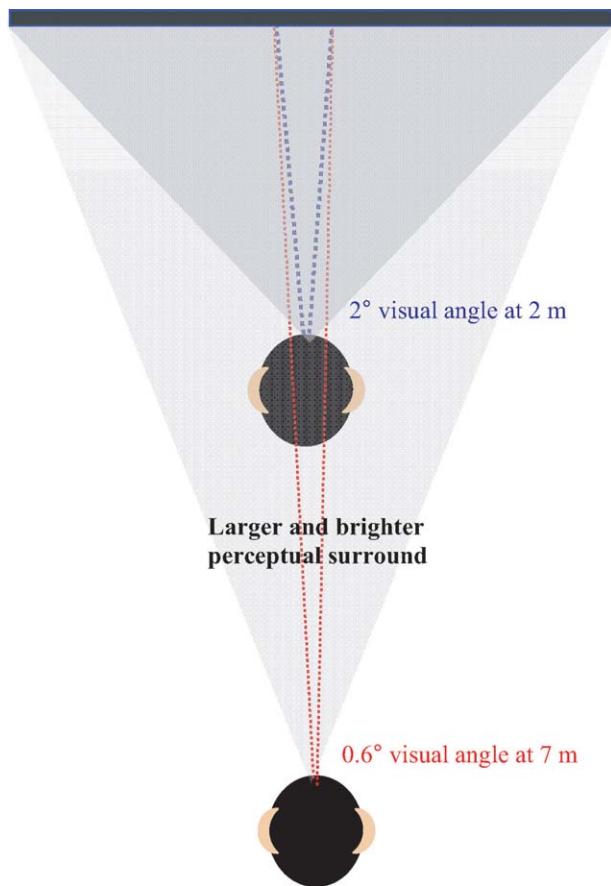


FIG. 9. Top view of DimGryIn2 and DimGryIn 0.6 phases. Angular display size is $\pm 13.1^\circ$ (H) $\times$ $\pm 7.6^\circ$ (V) at 2 m viewing distance and $\pm 3.8^\circ$ (H) $\times$ $\pm 2.2^\circ$ (V) at 7 m viewing distance.

colours to look less colourful but the medium and high chromatic colours to look more colourful.

Effect of Change in Background Luminance

Figures 11(a) and 11(b) plot the visual lightness and colourfulness results, respectively, of the black back-

ground (Phase DrkBlkIn2) against those of the gray background (Phase DrkGryIn2). Most of the test colours appear lighter against the black background than those against the gray background. The black background made most of the light colours less colourful than the gray background while the opposite trend was found for the dark colours. This lightness dependency of colourfulness change by different background luminance confirms the earlier finding by Hunt.¹⁵

COMPARISON OF THE MEAN VISUAL RESULTS AND CIECAM02 PREDICTION

The present visual results were used to evaluate the performance of the CIECAM02 colour appearance model.¹⁷ The results are summarized in Table III in terms of CV value. The luminance value of the adapting field, L_A , and the background luminance factor, Y_b , were calculated using measured data: $L_A = L_{DW} \times Y_b/100$, where L_{DW} is the luminance of a reference white on the display (in cd m^{-2}) and Y_b is the luminance factor of the background. For Phases BrtGryIn2, AvgGryIn2, AvgGryOut2, and BrtGryOut2, the ‘average’ surround parameters defined in CIECAM02 were used, the ‘dim’ surround parameters were used for Phase DimGryIn2 and DimGryIn0.6, and the ‘dark’ surround parameters for Phase DrkGryIn2, DrkGryIn0.6, and DrkBlkIn2. To compare the model predictions with the visual lightness and hue results, the original visual data were used. To compare the model predictions and visual colourfulness results, a scaling factor was first calculated using the least-squares method to adjust the model predictions onto the same scale as the visual results for each phase, because visual data and model predictions have different magnitudes. The average of the scaling factors, 0.87, of all nine phases was used to test CIECAM02 performance.

It can be seen in Table III that the CV value for Phase DrkBlkIn2 is much larger than those of the other phases. It was identified by Park *et al.*²² that CIECAM02 did not

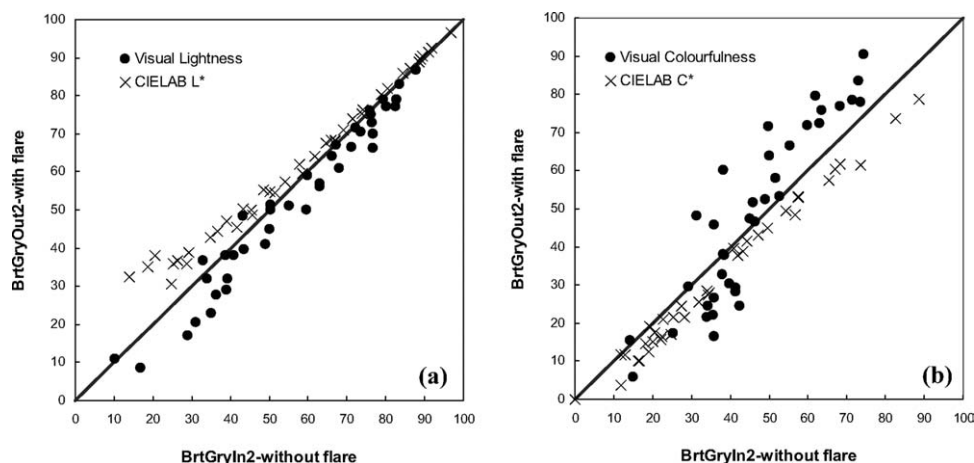


FIG. 10. Comparison of (a) the visual lightness (●) and the corresponding measured L^* data (x), and (b) the visual colourfulness (●) and the corresponding measured C^* data (x) of a series of test colours seen for BrtGryIn2, and the equivalent those seen for BrtGryOut2, respectively. Also shown is a line at 45 degrees.

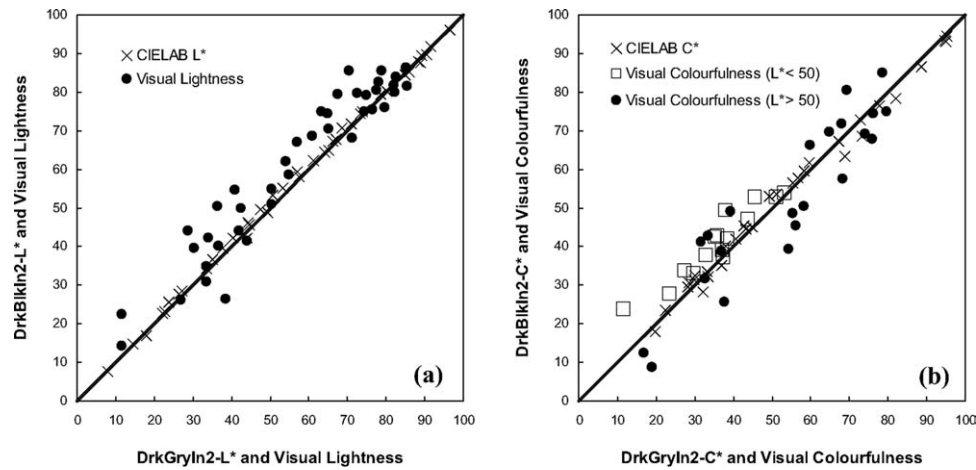


FIG. 11. Comparison of (a) the visual lightness (●) and the corresponding measured L^* data (x), and (b) the visual colourfulness (□ and ●) and the corresponding measured C^* data (x) of a series of test colours on the gray background (DrkGryln2), and the equivalent those on the black background (DrkBln2), respectively. The visual colourfulness data symbolized by squares are dark colours having L^* less than 50. Also shown is a line at 45 degrees.

perform well for the black background condition. Hence, their version of CIECAM02 was used: the original N_{cb} formula [Eq. (2)] was changed to Eq. (3). After the recalculation of colourfulness predictions for all phases, the CV values are again given in Table III (see “New N_{cb} ” column). A large improvement is seen for Phase DrkBln2.

$$N_{cb} = 0.725 \cdot (1/n)^{0.2} \quad (2)$$

$$N_{cb} = 0.725 \cdot (1/n)^{0.1425} \quad (3)$$

$$\text{where } n = Y_b/Y_w.$$

Figures 12(a)–12(c) show the plots of the mean visual data of DrkBln2 against the data predicted by CIECAM02 for lightness, colourfulness and hue, respectively. The comparisons between visual colourfulness and the predicted colourfulness before and after modifying N_{cb} are highlighted with the open circles and the filled squares, respectively, in Fig. 12(b). Colourfulness was over-predicted before the modification; however improvement can be seen after the modification i.e., the filled squares are distributed around the 45° line. In testing CIECAM02 lightness and hue as shown in Figs. 12(a) and 12(c), respectively, all data points are close to the 45° line, indicating good performance of CIECAM02 in predicting lightness and hue for colours against a black background.

The comparisons between the mean visual lightness and the lightness predicted by CIECAM02 are shown in Fig. 13 for all phases except DrkBln2 which has already been shown in Fig. 12. The abscissa and ordinate denote the predicted and visual data, respectively. Overall the lightness prediction by CIECAM02 is satisfactory, however all phases seem to have a general trend as shown by the broken curves, i.e., the very light and dark colours are located below the 45° line (an overprediction by CIECAM02). This could be due to an artifact of the magnitude estimation method. When observers assess lightness for the very light colours, they tend not to give the

extreme lightness scale values. For the medium light colours, most data points are positioned earlier the 45° line, i.e., visual lightness is larger than the lightness predicted by CIECAM02. This trend was also found when comparison was made between the visual data and those predicted by CIECAM02 using LUTCHI^{4,5} data set. However, the comparison plots are not shown here.

Figure 14 compares the mean visual colourfulness with the colourfulness predicted by CIECAM02. The data points look to be more scattered than those in Fig. 13. This is consistent with similar visual results elsewhere⁵ in which colourfulness has larger observer errors than the other two attributes as explained in Section 2–3. The CV values (19 – 23 in “New N_{cb} ” column in Table III) computed between the predicted and visual colourfulness are similar to those computed for observer accuracy (19) and observer repeatability (21) except for Phase BrtGryln2. Based on this comparison, it can be concluded that CIECAM02 already predicts visual colourfulness reasonably well, although there is some scattering of the data points.

Note that in “Effect of the Surround Excluding Glare” section, Phase BrtGryln2 showed the lightness dependency of colourfulness change by different surrounds, i.e., light colours appeared less colourful under the bright

TABLE III. Performance of CIECAM02 in terms of CV values.

Phase	Lightness	Colourfulness	Colourfulness (New N_{cb})	Hue
DrkGryln2	14.9	18.4	19.3	4.9
DimGryln2	11.4	21	20.6	6.1
BrtGryln2	19.5	24.5	25.7	7.4
AvgGryln2	18.9	18.9	18.8	5.2
AvgGryOut2	13.2	22.9	22.1	6.1
BrtGryOut2	15.9	23.7	22.3	6.4
DrkBln2	9.9	38.9	19.6	5.6
DrkGryln0.6	12.2	24.2	22.7	5.7
DimGryln0.6	13.8	21.5	21.1	6.5
Mean	14.4	23.8	21.4	6.0

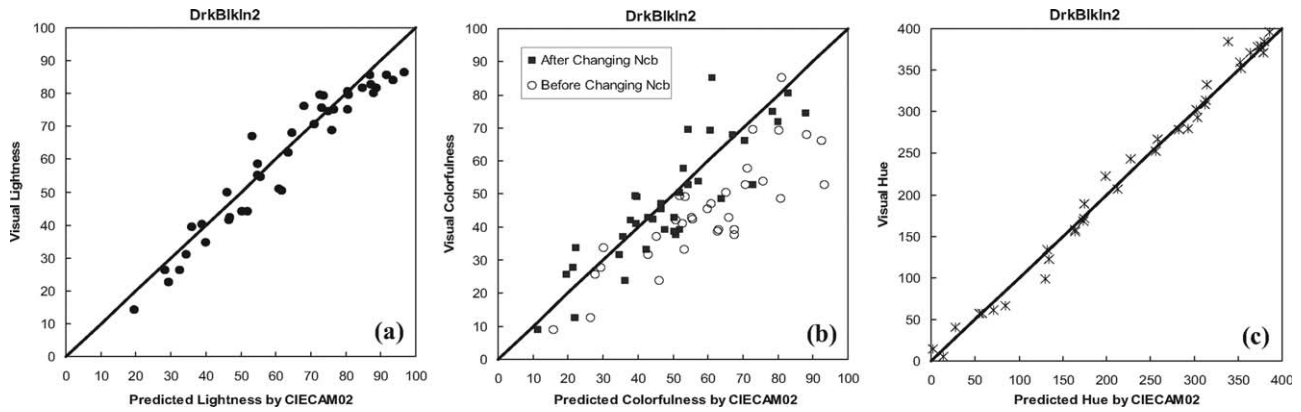


FIG. 12. Comparisons between (a) the mean visual lightness and the corresponding predicted lightness by CIECAM02, (b) the mean visual colourfulness and predicted colourfulness by CIECAM02, (c) the mean visual hue and predicted hue by CIECAM02, for the 40 test colours on the black background (DrkBIn2). For the colourfulness comparison in (b), open circles and filled squares are the comparisons before and after modification of N_{cb} respectively. Also shown is a line at 45 degrees.

surround than under the dark surround whereas dark colours looked more colourful under the bright surround than under the dark surround. CIECAM02 however does not model these effects, hence the CV value for BrtGryIn2 was the largest amongst the nine phases studied. Most light colours (L^* greater than 50) were over-predicted whereas dark colours were under-predicted by CIECAM02. Further psychophysical experiments are necessary to confirm this result.

For the comparison between the mean visual hue and the CIECAM02 prediction, all data points lay close to the 45° line, indicating good performance in hue prediction regardless of the changes in surround and background conditions. Hence, this data is not shown.

CONCLUSIONS

Nine phases of experiments were carried out to investigate how the change of surround conditions affected the colour appearance of test colours seen on a large display including one phase using a different background. The surrounds were divided into two groups. For the surround excluding glare, the illumination hardly affected the displayed colours because the level of glare light was very low. Observers, however, could perceive either a bright or a dim surround. For the surround including glare, the observers could see veiling glare added to the displayed colours with a bright surround. The second group represents typical viewing conditions which can be met in real life.

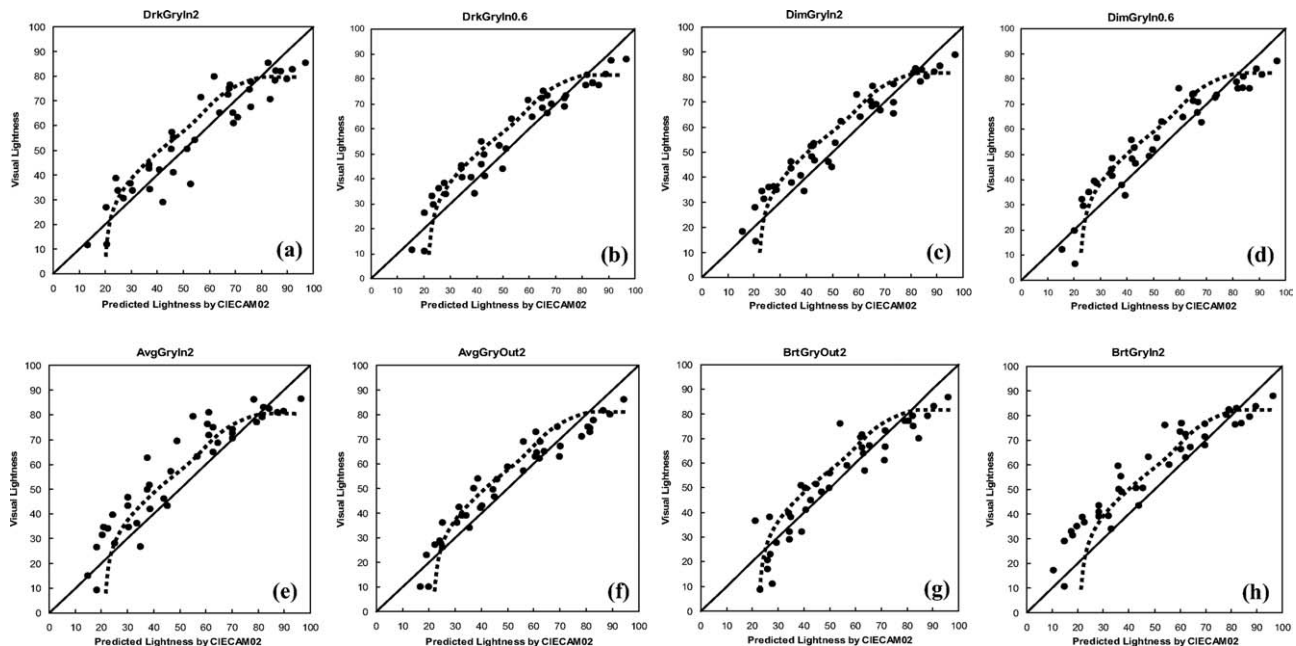


FIG. 13. Comparisons of mean visual lightness and the corresponding predicted lightness by CIECAM02 for (a) DrkGryIn2, (b) DrkGryIn0.6, (c) DimGryIn2, (d) DimGryIn0.6, (e) AvgGryIn2, (f) AvgGryOut2, (g) BrtGryOut2, (h) BrtGryIn2. Also shown is a line at 45 degrees.

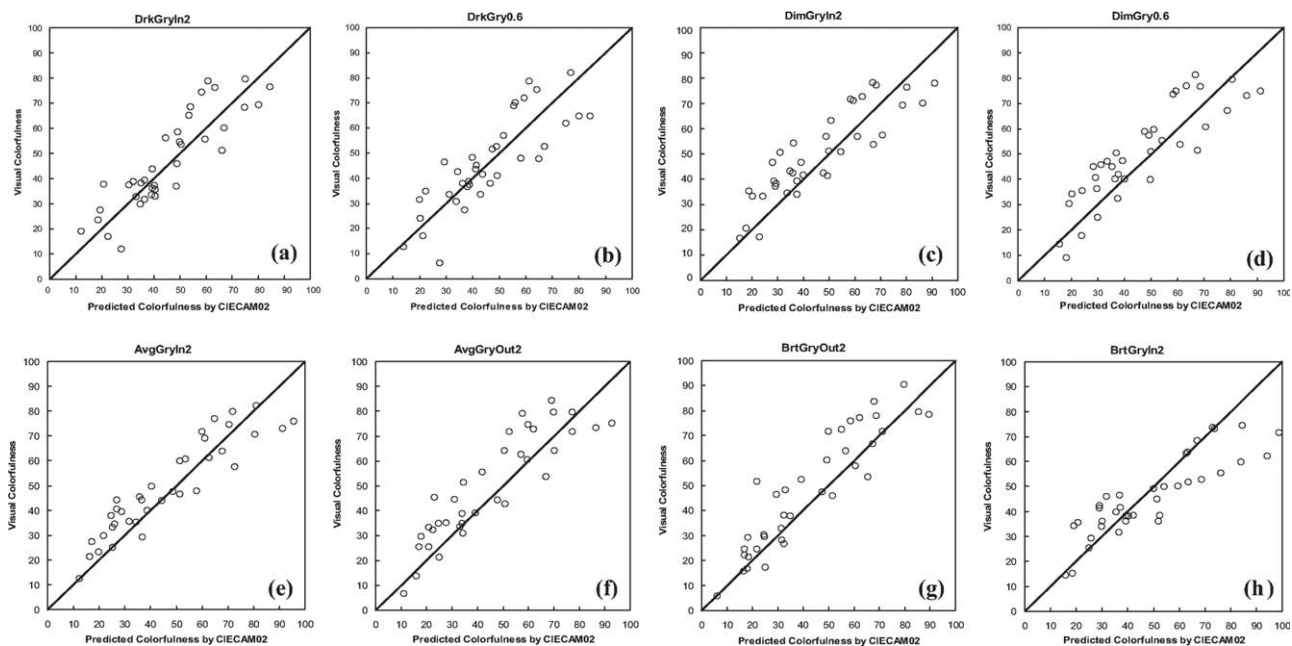


FIG. 14. Comparisons of mean visual colourfulness and the corresponding predicted colourfulness by CIECAM02 for (a) DrkGryIn2, (b) DrkGryIn0.6, (c) DimGryIn2, (d) DimGryIn0.6, (e) AvgGryIn2, (f) AvgGryOut2, (g) BrtGryOut2, (h) BrtGryIn2. Also shown is a line at 45 degrees.

For the effect of the surround excluding glare, the visual lightness results between the bright and the dark surrounds, and between the dim and the dark surrounds were almost identical. This seems to disagree with Bartleson and Breneman effect.⁹ The reason for this small difference might be due to that the difference in stimuli used between the current work and Bartleson and Breneman's study. The visual colourfulness results, however, showed lightness dependency.

For the effect of the surround including glare, the visual lightness and colourfulness results were compared with the CIELAB L^* and C^* values measured during each experiment. It was found that observers discounted the glare light consistently when judging the lightness of dark colours. For light colours, as the surround became brighter these were perceived as being darker. In the visual colourfulness results, the low chromatic colours appeared less colourful for the bright surround than for the dark surround, making it difficult to distinguish the shadow-detail in an image in bright outdoor conditions.

Additionally, the combined effect of changes in stimulus and surround sizes was investigated by varying viewing distance. There were no colour appearance differences between the different stimulus sizes, 2° and 0.6° , with a dark surround. However, with a dim surround, very dark colours looked less colourful and darker with the 0.6° stimulus size, i.e., with the longer viewing distance, due to the larger and brighter surround in the observers' field-of-view. This suggests that the appearance of colours presented on displays can be influenced by changes in both stimulus (image) and surround size.

Colour appearance data from the nine phases were used to test the performance of the CIECAM02 colour appearance model.

The colourfulness prediction with the black background was the worst and so the appropriate parameter (N_{cb}) in the model was adjusted to improve the predictions.

Further work has been carried out to verify the findings described here using complex colour images presented on the same display. The results will be reported subsequently.²⁰

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Publications Briefly Mentioned

CIE 185:2009 Reappraisal of Colour Matching and Grassmann's Laws, Vienna: CIE Central Bureau, 2009, 38 €, pp. 20–, ISBN 978 3 901906 78 7

The laws of additivity and proportionality of color matches, Grassmann's laws, are the basis of all color theory, but are not axiomatically true. The extent of departure of human vision from Grassmann's laws has been periodically examined. One exploration, W. A. Thornton found considerable failure of transformability of primaries—a symptom of Grassmann additivity failure. In the years since Thornton's finding, several groups have been formed to replicate and understand Thornton's results and the limitations of Grassmann's laws. CIE TC 1.56 is the latest of these. During the 10 years of this committee's existence, statistical simulations indicated that replicate matches by the same observer (not present in Thornton's data) are required to suppress random errors, and accordingly three laboratories generated intraobserver matching results in three different luminance

domains. Two of the studies conducted at 300 cd m^{-2} and 30 cd m^{-2} , respectively, confirm Grassmann additivity, but the third study shows failure of additivity at 3 cd m^{-2} . In addition, Maxwell and maximum-saturation color matches have long been known to be inconsistent even at high-luminance levels and with intraobserver match replication to suppress noise. A practical consequence of the failure of additivity could be problems observed in cross-media color matching, although cross-media studies also have other well-known sources of imprecision when the color-matching is asymmetric. Some suggestions are made for a covering theory of Grassmann's laws that might accommodate both Maxwell and maximum-saturation match data while still maintaining consistency with high-luminance success in experiments such as reported recently. Further investigations are indicated for a successor to TC 1.56.

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